Question 3: Develop and Monitor a Machine Learning-Based Online Payment Fraud Detection System

A digital payment company wants to detect fraudulent transactions and continuously monitor its deployed machine learning model for performance issues, errors, and changes in transaction patterns.

Using the online payment transaction dataset provided by the examiner, develop and monitor a Machine Learning-Based Fraud Detection System by performing the following tasks:

1. Load the transaction dataset, identify missing values, and perform the necessary preprocessing and feature engineering.

2. Train a machine learning classification model to classify transactions as Legitimate or Fraudulent and evaluate its performance using appropriate metrics.

3. Deploy the trained model using FastAPI, implementing /predict and /health endpoints.

4. Implement application logging to record incoming transaction requests, prediction results, processing time, and errors.

5. Integrate Prometheus to collect application metrics such as request count, prediction count, error count, and response latency.

6. Configure Grafana to visualize the collected metrics through an interactive monitoring dashboard.

7. Use Evidently AI to analyze data drift by comparing reference transaction data with newly generated transaction data.

8. Simulate changes in transaction amounts, transaction frequency, and payment methods, and demonstrate how the monitoring system identifies data drift and application performance changes.

9. Analyze the monitoring results and suggest appropriate actions to maintain model reliability.